

ASSIGNMENT – 07

Q7. Arduino enables interaction with real-world systems through Digital Pins, Analog Pins, and PWM-enabled Pins. Provide a detailed analytical discussion explaining how these three categories of pins collectively support sensing, decision-making, and control in embedded and IoT systems.

Arduino UNO R3 has many but here it has 3 types of pins:

- Digital Pins
- Analog Pins
- PWM Pins

1. Digital Pins:

Here digital pins are nothing they just work as a light and switch. And they are the most basic interface, operating on discrete binary logic.

- **Signals:**

They represent data in two states: HIGH (typically 5V or 3.3V) and LOW (0V/Ground). These correspond to Boolean true and false.

- **Modes of Operation:**

- **INPUT:** Sets the pin to a high-impedance state, allowing it to sense external voltages.
- **OUTPUT:** Allows the pin to provide current to a circuit.

- **Components:**

- **Output:** LEDs, buzzers, or relays.
- **Input:** Pushbuttons, switches, or PIR motion sensors.

- **Practical Consideration:**

Where we use the Arduino uno r3 board to blink an external LED with its onboard LED. So first we connect the LED cathode to GND(Ground) Pin and anode to Digital pin (12), then we write the code on Arduino sketch platform after deploying the code the LED will start blinking.

2. Analog Pins:

Here analog pins handle the flow of Voltage.

- **Signals:**

These pins measure voltage between 0V and the reference voltage (usually 5V).

- **Role of ADC:**

The Analog-to-Digital Converter translates voltage into a digital number. Arduino usually uses a 10-bit ADC, mapping 0V–5V to a range of 0 to 1023.

- **Components:** Potentiometers, LDRs (light sensors), thermistors (temperature), and moisture sensors.

3. PWM pins:

PWM pins are nothing they are also digital pins but it simulates the analog behavior through rapid switching. It means if u want to control the voltage or how much power something gets, plug it into a pin with (~) this symbol.

- **Concept:**
By switching a digital signal ON and OFF very quickly, PWM creates an "average" voltage. The Duty Cycle (the percentage of time the signal is HIGH) determines the perceived output level.
- **Analog-like Control:**
While the signal is technically digital, the fast switching allows for dimming LEDs or varying motor speeds without a true analog output.
- **Components:**
DC motors (speed control), RGB LEDs (color mixing), and Servo motors (positioning).

4. Integrated Analysis:

Feature	Digital	Analog	PWM Pin
Use case	Simple switching	Sensing environment	Variable control/voltage control
Direction	Input/output	Input	Output
Signal type	Binary (Discrete)	continuous	Simulated analog

5. Work on Real world:

In a real-world system (e.g., an automated greenhouse), these pins work in a loop:

- **Sensing:**
Analog pins read soil moisture levels.
- **Processing:**
The Arduino logic decides if the plant needs water.
- **Actuation:**
Digital pins trigger a relay to turn on a pump, while PWM pins might adjust the speed of a cooling fan based on temperature.

6. Engineering Trade-offs:

Choosing between pins requires balancing resolution vs. speed. While Analog pins provide rich data, they are slower to read than Digital pins. Furthermore, engineers must consider pin multiplexing (sharing pins) and power consumption when designing complex IoT systems